Chapter 1 · Energy Causes Change

GUIDED NOTES

Name: _____ **Homeroom:** ○ 4A ○ 4B ○ 4C ○ 4D

BQ Where can we observe evidence of energy causing change?

My first thinking

_____

_____

V1 VOCABULARY – ENERGY & CHANGE

Reader p. 1

energy – the ability to cause _____ .

change – to become _____ .

N1 ENERGY CAUSES CHANGE

Reader pp. 1–2

Energy has the ability to cause change.

When you see a change, you are seeing _____ of energy.

ENERGY → **CHANGE** →

N2 ENERGY OCCURS IN DIFFERENT FORMS

Reader p. 3

Write one example that shows a change for each form.

FORM	ONE EXAMPLE THAT SHOWS A CHANGE
motion	
sound	
light	
heat	
electrical energy	

V2 VOCABULARY – STORED ENERGY

Reader p. 4

stored energy – energy that has the ability to cause change at a _____ time.

N3 OBJECTS CAN STORE ENERGY

Reader pp. 4–5

STORED ENERGY → **RELEASE** → **CHANGE**

EXAMPLE	WHAT CHANGE COULD HAPPEN LATER?
battery (in a flashlight)	
stretched rubber band	
raised object (apple)	

Objects also have stored energy because of their _____ .

N4 RECOGNIZING EVIDENCE

Reader pp. 1–5

Draw a change you observed. Show what it looked like BEFORE and AFTER.

BEFORE	AFTER

What changed?

CHECK YOUR THINKING

How do we know energy is involved if we cannot see energy itself?

Start with: "We know because we see …"

BQ · NOW I THINK Where can we observe evidence of energy causing change?

Practice — Energy Causes Change

Chapter 1 · Grade 4 · PRACTICE — NOT GRADED

Name: _____ **Homeroom:** ○ 4A ○ 4B ○ 4C ○ 4D

A WHAT CHANGED?

Write the change you could observe and one form of energy.

SCENARIO	WHAT CHANGED?	FORM OF ENERGY
A light switch is flipped.		
A toy car is pushed.		
A bell is rung.		
An ice cube sits on a warm hand.		

B EVIDENCE OR NOT ENOUGH EVIDENCE?

Circle one. Then finish the sentence: "Because …"

OBSERVATION	CIRCLE ONE		BECAUSE …
The fan blades started spinning when we plugged it in.	**EVIDENCE**	**NOT ENOUGH**	
The battery looks shiny.	**EVIDENCE**	**NOT ENOUGH**	
The room got brighter when the light came on.	**EVIDENCE**	**NOT ENOUGH**	
I think the motor is strong.	**EVIDENCE**	**NOT ENOUGH**	

C STORED ENERGY

Circle YES or NO. Then tell what change could happen later — or say if it is happening now.

EXAMPLE	STORED ENERGY?	CHANGE LATER — OR HAPPENING NOW?
a battery in a flashlight that is switched off	YES NO	
an apple held above the ground	YES NO	
a stretched rubber band	YES NO	
a toy car rolling across the floor	YES NO	

D EXPLAIN

Energy is the ability to cause change. Describe one change you have observed. Tell why that change is evidence of energy.

Chapter 1 Quiz — Energy Causes Change

Chapter 1 · Grade 4 · SCORED — 9 questions · 16 points

Name: _____ **Homeroom:** ○ 4A ○ 4B ○ 4C ○ 4D

Notes today: ○ **OPEN NOTE** ○ **CLOSED NOTE** *(teacher circles one)*

SCORE: _____ **/ 16** _____ **%**

1. Energy is the ability to … *(1 pt)*

a) be invisible
b) disappear
c) cause change
d) make sound

2. When you see a change, that change is … that energy is present. *(1 pt)*

a) not important
b) evidence
c) impossible
d) a guess

3. A student pushes a toy car and it rolls across the floor. *(2 pts)*

What changed?

_____

_____

What form of energy do you observe?

_____

4. Which list names forms of energy? *(1 pt)*

a) red, blue, green
b) up, down, left, right
c) fast, slow
d) motion, sound, light, heat, electrical energy

5. Fill in the blanks to show the model we use. *(2 pts)*

Word bank: evidence change

ENERGY → _____ → _____

6. A battery is sitting still in a flashlight that is switched off. Does the battery have energy? Explain. *(2 pts)*

7. A book is held up high above the floor. It is not moving. The book ... *(1 pt)*

a) has stored energy

b) has no energy

c) is moving

d) is lighter

8. A student flips a switch and the light bulb turns on. Is this evidence of energy? Explain. Use the word "change." *(3 pts)*

9. Circle YES or NO: is it evidence that energy caused a change? *(3 pts)*

STATEMENT	CIRCLE ONE	
a) The flag waved when the wind blew.	**YES**	**NO**
b) The desk is brown.	**YES**	**NO**
c) The music got louder.	**YES**	**NO**